Socialnetwork Diffusion with LLMs
Technical Report

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1 Data Interpretation [Banerjee et al., 2013]

A trivial feature analysis based on covariance and variance coverage reveals that household covariates are not obvious, strong predictors of adoption and transmission rate in the typical machine learning sense. However, in case of LLMs, we may still leverage the demographic information as wealth proxies (`rooms`, `beds`, per-capita variants, `electricity`, `own_latrine`, `occupation_head`), exposure/attitude to financial services (`has_shg`, `has_savings`), and in-group/out-group relations (`subcaste` and `religion`). In this variant, we consider `leader` as an implied feature, rather than an explicit public attribute. This is follows from how the "leaders" were chosen based on their occupation, from the intuition that this feature would have a dominant influence when fed directly to a language model in the adoption phase[1]. However, `leader` is incorporated as a conditional feature in the transmission phase, following from the paper's description of the seeding routine. Finally, `adopted` also becomes a conditional attribute in the transmission phase. Everything else is dropped: redundant or superseded fields, fields missing in high ratio of records, fields redacted at source, near-constant fields, and bare or vague identifiers.

This study considers the above feature sets not as direct predictors of adoption or transmission ratios, but as foundational traits for grounding the narrative capacity and heterogeneity of the LLMs in plausible, realistic context.

Edge type In particular, while the type of edge seem like potential predictors and narrative features, we choose not to use them in this variant. Nothing in the survey records who told whom, and the survey didn't ask for justifications. There is no ground truth for which of the twelve relationship types actually carried information or influence. Similarly to the original paper's implementation, here `informed` is a state, rather than a feature for inference. Consequently, the simulation also runs on the union network rather than trying to select or weight by edge type.

2 Agent-Based Model

We take a decision-theoretic approach and seperate the mechanics into two simple "games".

Transmission Every `informed` household i, in each turn, for each (not-yet-adopted or not-yet-informed) neighbour $j \in N$ plays the transmission game $G^T = \langle A, S, P_i, U_i \rangle$, where the action space – in the context of the *Will this household inform the neighbour?* decision scenario – is $A = \{\text{YES}, \text{NO}\}$, the

[1] Something to be properly investigated in a full project....

states of nature are $S = \{\text{THEY JOIN}, \text{THEY DON'T JOIN}\}$, and the subjective utilities U represent a $\frac{\text{risk}}{\text{reward}}$ relation for the given agent[2]. Finally, P_i are the probabilities that the agent associates to each state of nature. This model builds on the assumption that information transmission is not a neutral act – e.g. [Crawford and Sobel, 1982] –, and the consequences of transmission will yield utility similar to a coordination game[3]: *If I decided to join and telling them about it, I influence my neighbour to join, I will experience satisfaction.*

Adoption Whenever a household i gets `informed`, it plays the adoption game $G^A = \langle A, S, P_i, U_i \rangle$ where the action space – where the question is *Will this household adopt the microfinance services?* – is $A = \{\text{YES}, \text{NO}\}$, the states of nature pertaining to the potential effects on the household's financial circumstances are $S = \{\text{BENEFICIAL}, \text{LIMITED}, \text{HARMFUL}\}$, and the subjective utilities U represent a $\frac{\text{risk}}{\text{reward}}$ relation for the given agent. Here, we assume that humans will weigh the potential personal financial consequences of having access to loans, which, at a high level and time-agnostically, can be modelled in the simplest way as positive (e.g. it helps pay for house improvements), neutral (e.g. it is used as a balance transfer), or negative (e.g. the loan is misused and results in a debt spiral). As before, P_i are the probabilities that the agent associates to each state of nature.

3 Agent Design

Agents are households, not individuals: the individual survey covers only median 46.8% of a village, albeit non-randomly. We focus on households and household-aligned features.

A Note on Implementation I prefer to use strongly structured (as per old-school Wooldridge) agent architectures wrapping LLMs, rather than assuming inherent agency off-the shelf. This specific dataset and use case would nicely align with the Observe–Orient–Decide–Act cycle I like to use where we can model cognitive and physical features, and account for environmental uncertainties. While the sequential model I implemented still follows this concept and the round protocol (receive, decide, advance) is loosely OODA-shaped, this is not implemented here due to the time and complexity overhead. Design choices here are limited to the use of hybrid agents that mix the social context processing abilities of LLMs with the clibration of the algorithmic models presented in the paper or the purely synthetic autonomous LLM population[4]. Secondly, we investigate a narrow selection of prompt designs covering ideas from information sets, cognitive theory and decision theory as follows, for the adoption phase:

Ego profile (A) :

- **0** : No household info involved.
- **1** : Data incorporated as a listing.
- **2** : Data incorporated as an engaging narrative story that was generated by an LLM before hand and fixed per household id.

Neighbour profile (B) : Same as above.

Endorsement (C) :

[2]Here using the network edge types would likely carry influence that stems from the personal relation of the agents.

[3]Given a wider scope, I would have tested a version where the action space entails intentional influence: the agent can decide to endorse, inform, oppose, or ignore.

[4]However, even here, we take simplifying assumptions from the original authors to save on resources and runtime: agents are informed only once, agents decide only once, a single transmission is processed by each recipient at a time.

0 : What the neighbour decided is not mentioned.

1 : The neighbour's decision is mentioned.

Response format (D) :

0 Plain: Vanilla prompt.

1 Model of appropriateness: The agent answers 1. What kind of situation is this? 2. What kind of person am I? 3. What should someone like me do in a situation like this?

2 Decision-theoretic: The LLM must provide a structured JSON schema, populating a decision matrix according to the game definitions, providing a justification for each selected value.

The transmission phase follows the same approach, where C is replaced by a conditional parameter L(eader), used for the seeding prompts.

4 Experiment Design

Village 6 is selected as the initial target subset based on capita and edge count, not on data quality. At a five-round horizon its 730-edge network needs 3,650 edge-round calls against 7,824 (village 47) and 12,132 (village 73). The trade-off is the lower survey coverage of households and a flat trimester-based trajectory.

Timing artefact: q_N, q_P were fit to reproduce the eventual adoption pattern, not a documented sequence of conversations. While we adopt the rounds = trimester presence count assumption from the papers, the household-level adoption record and the panel's trimester curve are two different empirical objects, so simulated trajectory is to be viewed as a modelling artefact (albeit interesting!) rather than objective measure of human-likeness in this study.

LLM: This study used `gpt-5.4-nano-2026-03-17` as a starting point. It is the cheapest of the recent *SOTA* API models[5], fast in runtime, and, according to my experience with game-theoretic experiments, "smaller" models tend to be more human-like in their performance – e.g. in [Trencsenyi et al., 2025].

4.1 Pilots

Due to the cost and time constraints, we first conduct a pilot on both games to narrow down agent selection for a more extensive simulation. A search was conducted with $n = 25$ repetitions with each parameter combination and using carefully selected real samples from village 6.

Adoption Rates The search is scored on whether an LLM decision tells apart a real adopter from a real non-adopter (Figure 1). An interesting highlight beside best performers is that only two non-endorsed configurations discriminate between samples, this implies that the agents tend to pick up more from the influence of the neighbours' decision than from the demographic data[6]. In addition, as Figure 2 depicts, inconsistencies appear when reasoning is forced through a semi-formal bottleneck.

[5] Models after 5.5 default to "reasoning" under the hood, while older 5.x models' hidden reasoning can't be reliable turned off

[6] Based on this sample set. This requires an in depth analysis

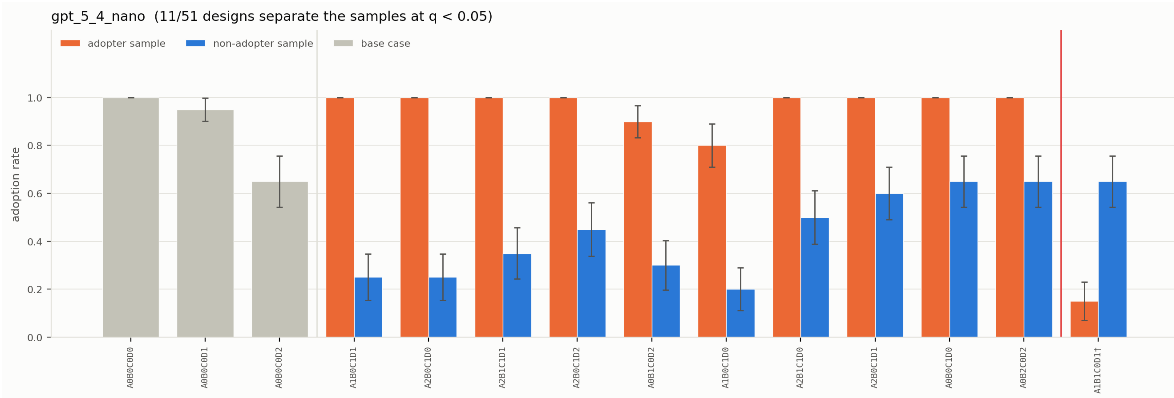

Figure 1: Adoption rate by design, filtered by statistical significance of discrimination, adopter vs. non-adopter sample household, $n = 25$ reps/arm. Grey: base cases. Red line: designs that invert.

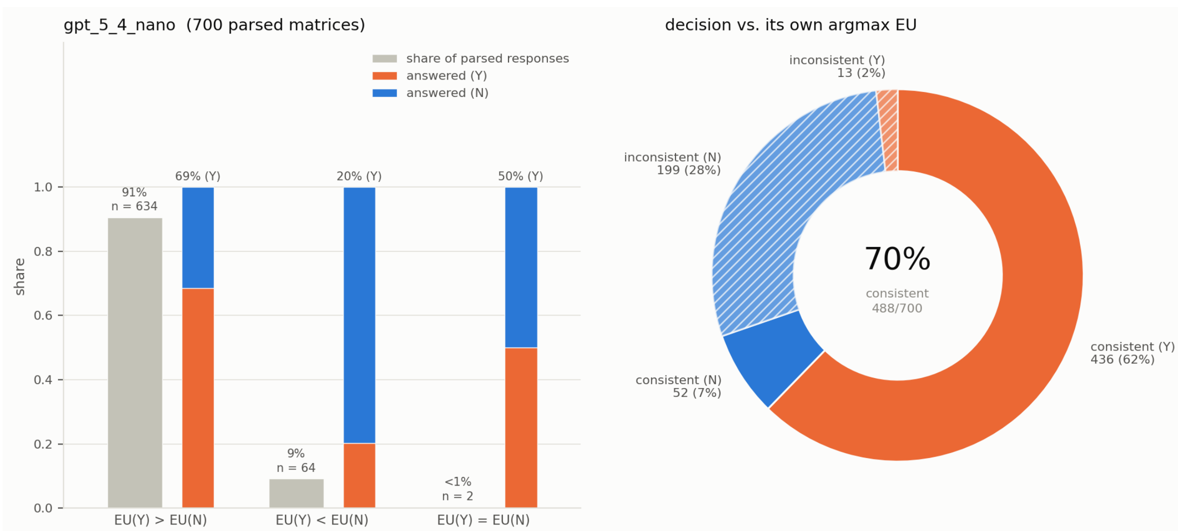

Figure 2: Rationality and consistency of LLM reasoning via decision theory in the adoption pilot.

Transmission Rates We apply the same approach as above, focusing in transmission rates. Lacking ground truth, Figure 3 can be compared to findings from the paper. While some models approximate the algorithmic prediction, Figure 4 indicates a high level of inconsistency [7].

Costs See Table 1.

Axis	Adoption (in / out)	Transmission (in / out)
A0 / A1 / A2	448/252 · 531/287 · 910/283	459/234 · 542/222 · 888/239
B0 / B1 / B2	458/264 · 554/277 · 877/281	450/222 · 541/232 · 899/240
C0 / C1 | L0 / L1	625/274 · 634/275	627/230 · 632/233
D0 / D1 / D2	465/97 · 505/255 · 918/470	501/126 · 542/183 · 847/385

Table 1: Adoption and transmission counts across experimental axes.

[7] Something our logical enchancement frameworks can aid with, e.g. Mensfelt et al. [2025]

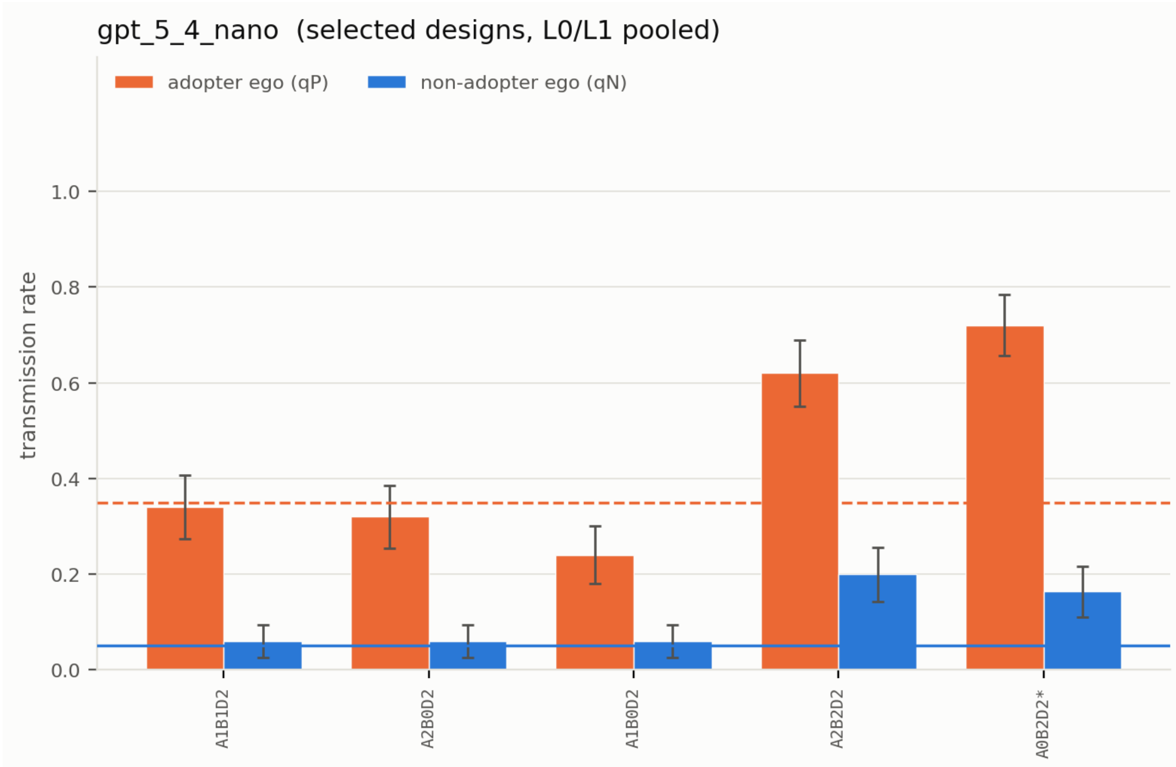

Figure 3: Transmission rate by design, filtered by statistical significance of discrimination, adopter vs. non-adopter sample household, top 5 lowest rates. $n = 25$ reps/arm.

4.2 Hybdrid model

We implement the paper's contagion algorithm to simulate the information transmission, while the LLMs take over for the decision-making on the adoption game. On the other side, LLMs are used to propagate the information, while adoption rates are replicated by the paper's model. The intention with this step is an ablation study that continuously places the agents into more complex situations, thus, informing model development and guiding model choice pruning for the more costly consecutive simulations. In this study, however, it was not completed due to lack of time. Instead, pilot and preliminary results, empirical observation, and intuition guided further modelling choices.

Algorithmic Transmission with LLM Decision-makers Preliminary poor results suggest a mismatch between the mechanism transfer in social-context-grounded-sampling-to-LLM or LLM-to-LLM vs. algorithmic-transmission-to-LLM. What seems to work in one settings, fails in the other.

LLM-driven Diffusion with Algorithmic Adoption `raise NotImplementedError()`

4.3 Synthetic LLM-driven Population

Extended by additional newer and older models. Reasoning models heavily overestimate adoption rates (Figure 5) and transmission rates (Figure 6).

Extension To test out-of-sample-potential, the same configuration was run on **Village 73**, that involves higher capita and a denser network. In addition, while the average case entails lower adoption rates from "leaders", in this village this feature is flipped, providing us an additional axis for

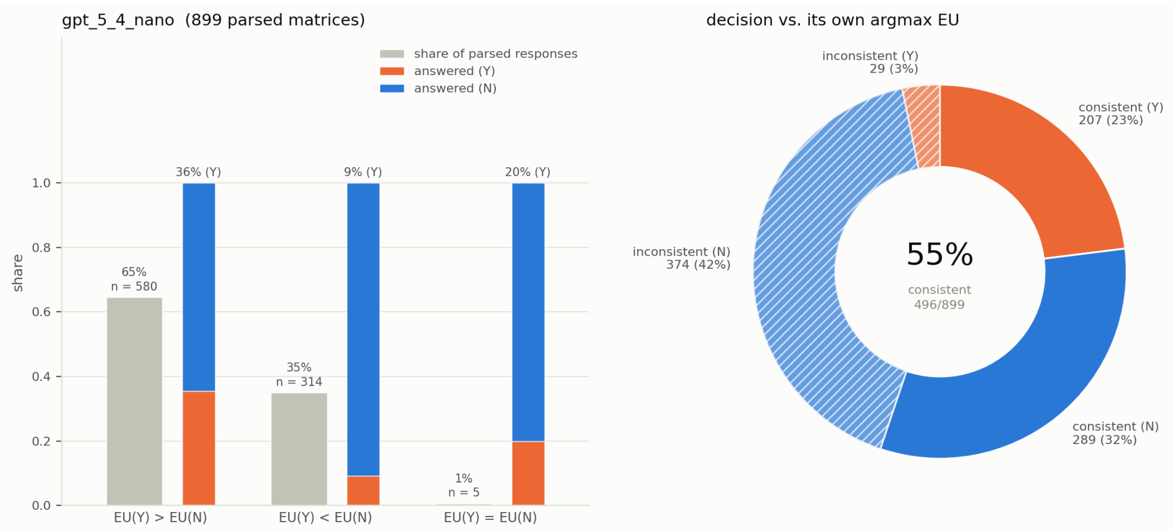

Figure 4: Rationality and consistency of LLM reasoning via decision theory in the transmission pilot.

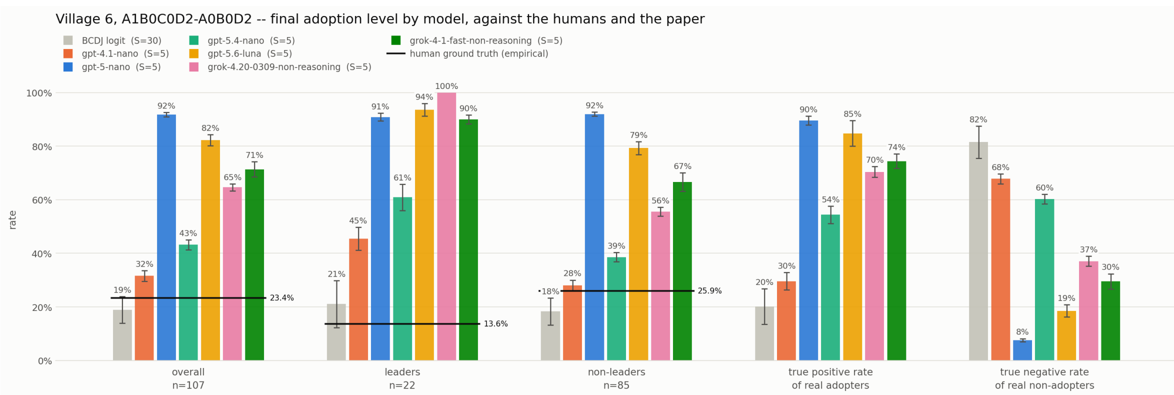

Figure 5: Village 6 adoption rates for fixed design for comparing models.

evaluating whether the responses are getting grounded in the data. – see Figures 7- 8.

In addition, we include alternative openai and xai models to test whether the "smaller-better" hypothesis transfers from game-theoretic experiments, and to evaluate how different models behave with the same configuration[8]. We focus on "non-reasoning" models to avoid conflating their hidden processes with our prompt designs' effects and to avoid incurred cost explosion. Note, Claude models are not covered as they don't support the same JSON schema through the openai API, and, older non-reasoning models are no longer available.

5 What does an LLM buy us here?

This study tested whether replacing fixed probabilities with context-sensitive artificial households improves our ability to reproduce how microfinance spread through a real village network, introducing individual-based decision-models and social-context-driven heterogeneity. However, current early results show potential sensitivity to social context, but not yet a demonstrated advantage over simpler

[8]This is an another entry for an in-depth analysis.

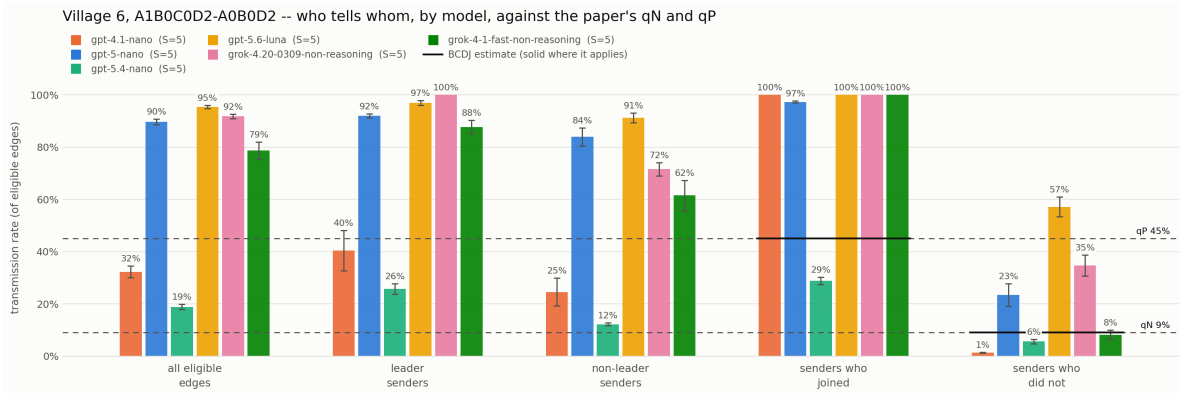

Figure 6: Village 6 transmission rates for fixed design for comparing models.

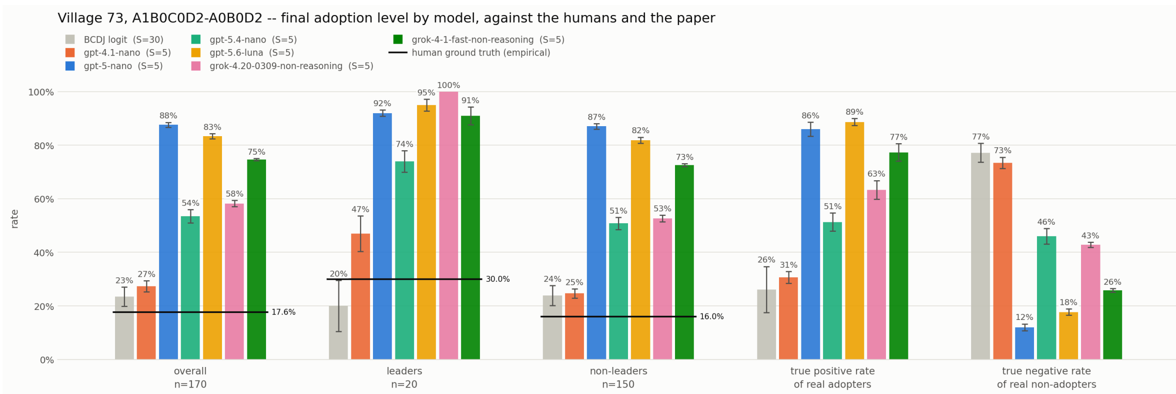

Figure 7: Village 73 adoption rates for fixed design for comparing models.

models.

In the optimal case, where the agent is properly calibrated and implements the appropriate verification methods, we can claim that the produced natural language or formal reasoning traces are plausible explanations a human may give in the same circumstances. While stronger claims can only be made in presence of reasoning ground truth, the plausible explanations may capture empirical mechanisms and latent variance stemming from the social context that algorithmic solutions cannot model.

The pilots and the single village experiments highlight the potential, but also show inconsistencies in the produced reasoning samples and consequent choices, which indicates a weaker grounding effect in the demogrpahic data.

6 A glance on potential stretch targets

The demographic-plus-network-plus-adoption structure here has cousins in experimental economics with WEIRD-population demographic data, a possible route to a Western comparison population via public goods games, and signalling games, or other canonical multi-stage-game sequential interaction models. For a feel on potential general adaptability and transferability, see e.g. [Binz et al., 2024].

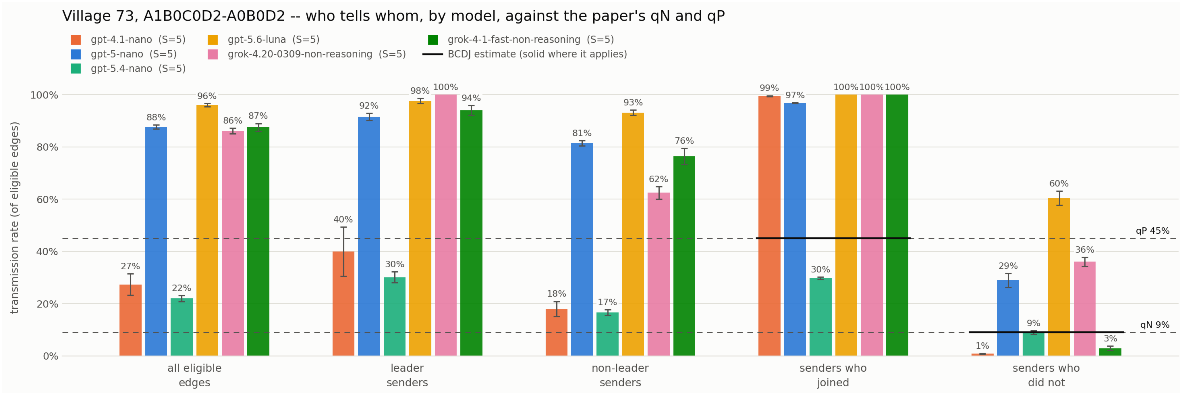

Figure 8: Village 73 transmission rates for fixed design for comparing models.

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